EDA Summary

The exploratory data analysis showed that the dataset is generally of high quality. It contains **35,040 observations** and **11 original features**, with no missing values or duplicate records. During outlier detection using the IQR method, **328 outliers** were identified in the Usage_kWh column. These outliers likely represent periods of unusually high energy consumption rather than data entry errors, so they were retained for analysis. While creating the Power_Factor_Ratio feature, one row produced an undefined value because both the leading and lagging power factors were zero. This issue was handled during preprocessing to ensure the dataset was suitable for modeling.

The correlation analysis revealed that **CO2(tCO2)** has the strongest positive correlation with Usage_kWh, followed by **Lagging_Current_Reactive.Power_kVarh. This indicates that higher electricity consumption is closely associated with increased carbon emissions and reactive power usage. Although the engineered High_Load feature also showed a high correlation, it was excluded from interpretation because it was directly derived from the target variable.

The visualizations showed a clear relationship between **Load Type** and energy consumption, with **Maximum Load** periods consuming significantly more energy than Light and Medium Load periods. The hourly energy usage plot also indicated that electricity demand varies throughout the day, suggesting that production schedules influence energy consumption.

Based on these findings, a reasonable hypothesis is that energy spikes occur during periods of intensive industrial activity when heavy machinery operates under Maximum Load conditions, leading to higher electricity usage, reactive power consumption, and CO_2 emissions.